Muhammad Athar

+92 326 9552003 | mathar.bese22seecs@seecs.edu.pk | linkedin.com/in/muhammad-athar-51450524a | github.com/AtharRizwan

Education

National University of Sciences & Technology – NUST — **CGPA**: 3.94
Bachelors of Engineering in Software Engineering — *Sept. 2022 – Present*

Experience

Deep Learning Research Intern — Feb. 2024 – Present
TUKL-NUST R&D Center — *Islamabad, PK*

- Performed literature review to explore the architectures that have been used on the assigned task.
- Gained expertise in pre-processing and training models on datasets as large as 60 GBs.
- Reproduced the research papers that yielded the best results, replicating the methodologies to understand their success.
- Explored new architectures to achieve a significant improvement in accuracy, enhancing performance by as much as 10%.

Projects

Scored 2.0 | *Python, NumPy, Matplotlib, Pandas, Tensorflow, Git* — Aug. 2023 – Sept. 2023

- Recurrent Neural Networks(RNNs) for ball-by-ball prediction of T20 cricket match scores and win percentages.
- Trained the models using data from 1773 cricket matches, which was obtained from the CricSheet database.
- Achieved an accuracy of 90.16% in predicting outcomes within 20 runs during the final 10 overs of the match.

Search Engine | *Python, React, Flask, Git* — Nov. 2023 – Dec. 2023

- Developed the project using 150,000 articles from the NELA-GT 2022 dataset, ensuring a comprehensive data source.
- Delivers search results within a second, even for lengthy or complex queries.
- Provides AI generated images based on prompts using Stable Diffusion Pipeline v0.24.0.

EEG Classification | *Python, NumPy, Tensorflow, Pytorch, Git* — Feb. 2024 – Present

- The project focuses on the classification of EEG signals into normal and abnormal.
- Created a pipeline for effectively managing different datasets and training various models.
- Recreated the results of a research paper that implements the SCNet architecture.
- Proposed an architecture for EEG classification task that achieves an accuracy of 90.14% on the NMT dataset.

Extracurriculars

Module Head Elevatech (Programming Competition) — May 2024
AIESEC IN NUST — *Islamabad, PK*

- Curated the problems for the competition, and authored a detailed guideline booklet to assist participants.
- A total of 39 teams, comprising 78 participants, actively took part in the competition.
- Supervised the competition closely to ensure fairness, taking measures to prevent any instances of plagiarism or misconduct.

Event Head Code o'Clock (A data science workshop) — Nov. 2023
ACM NUST Student Chapter — *Islamabad, PK*

- Oversaw the marketing strategies and financial operations for the workshop, ensuring effective promotion.
- 31 people participated and learnt useful things regarding data science from experienced people.

Certificates

Machine Learning Specialization — https://coursera.org/share/ddc6062cfc913fe02b10562731437bfc
DeepLearning.AI

Deep Learning Specialization — https://coursera.org/share/554f6e0d38eb1ec2907db88ed58eaa33
DeepLearning.AI

Natural Language Specialization — https://coursera.org/share/337b73ace9d951b50ffd3c1dd876a055
DeepLearning.AI

Technical Skills

Languages: Python, C/C++, HTML, CSS, JavaSript
Frameworks: React, Node.js, Django, Flask, Tensorflow, Pytorch
Developer Tools: Git, Docker, VS Code, Vim
Libraries: Pandas, NumPy, Matplotlib